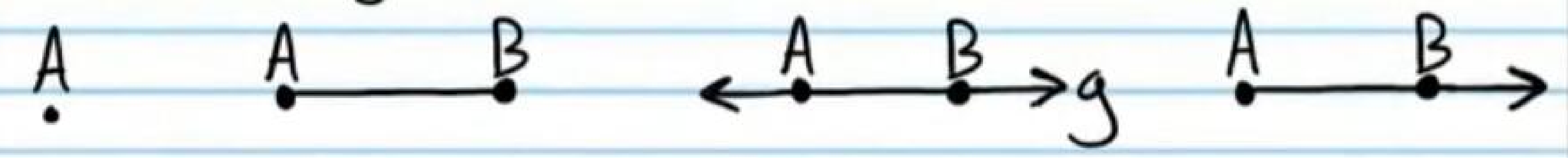
Name
Date
Course

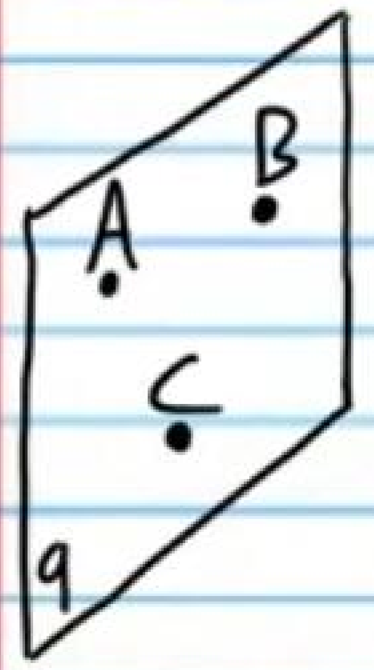
Fundamental Geometry (Part I)

Point | **Segment** | **Line** | **Ray**



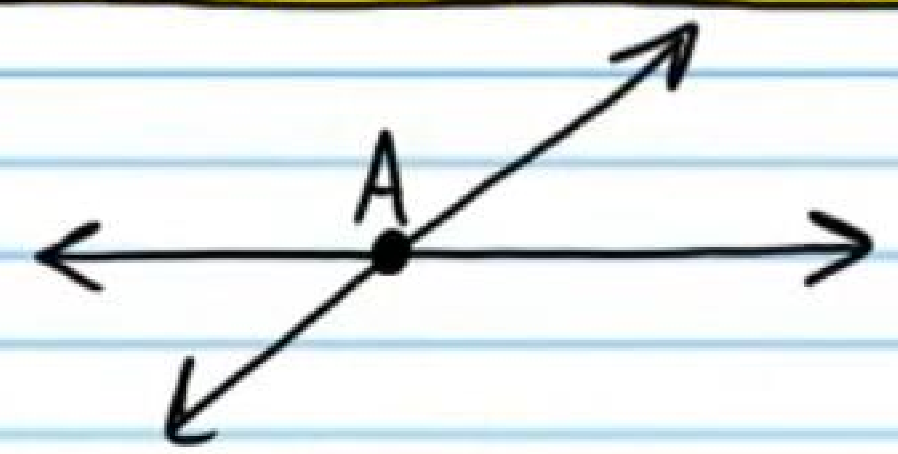
A | $\overline{AB}$ | $\overleftrightarrow{AB}$ or a lowercase letter such as "g." | $\overrightarrow{AB}$

Plane



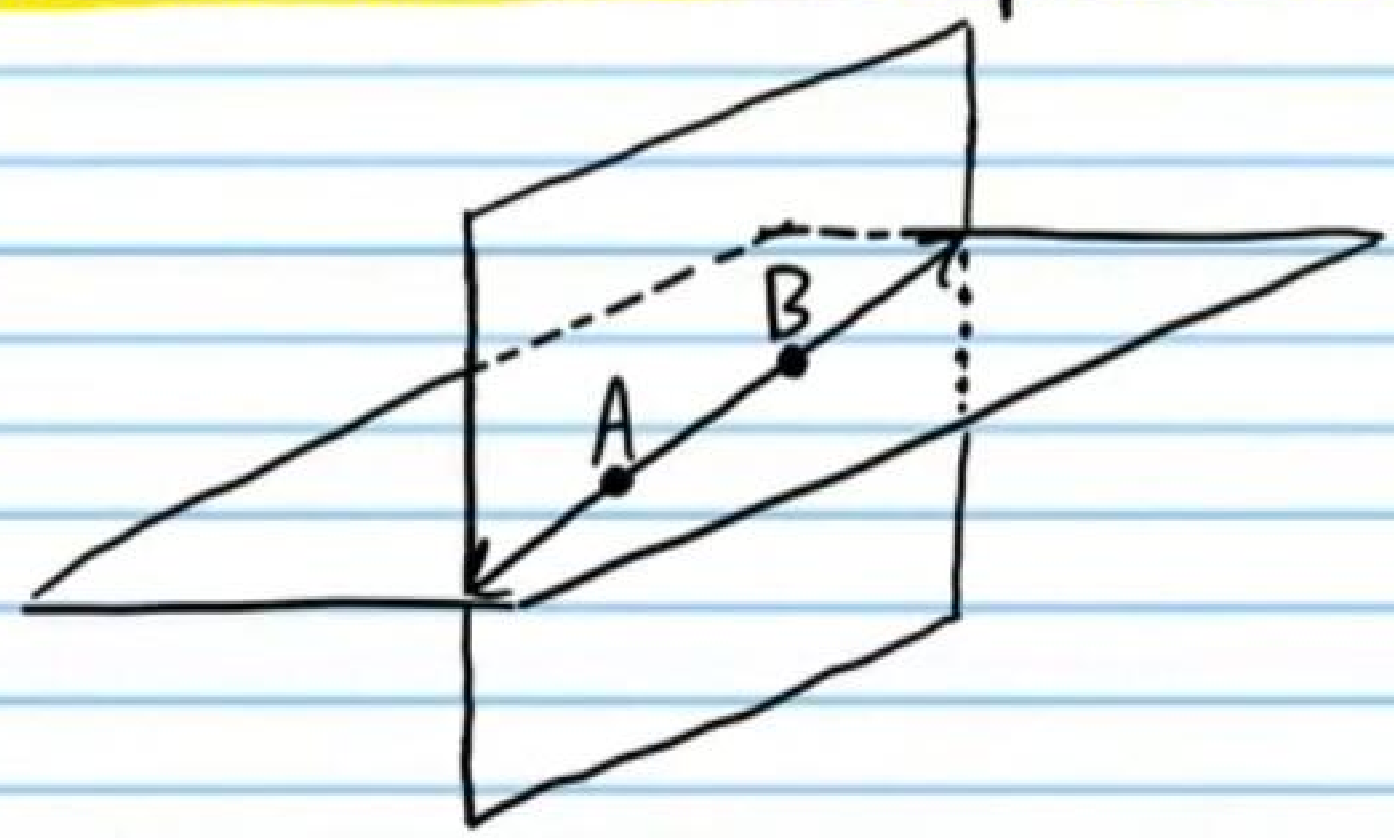
ABC or a lowercase letter such as "q."

The intersection of two lines is a point.



"A" is the point of intersection.

The intersection of two planes is a line.



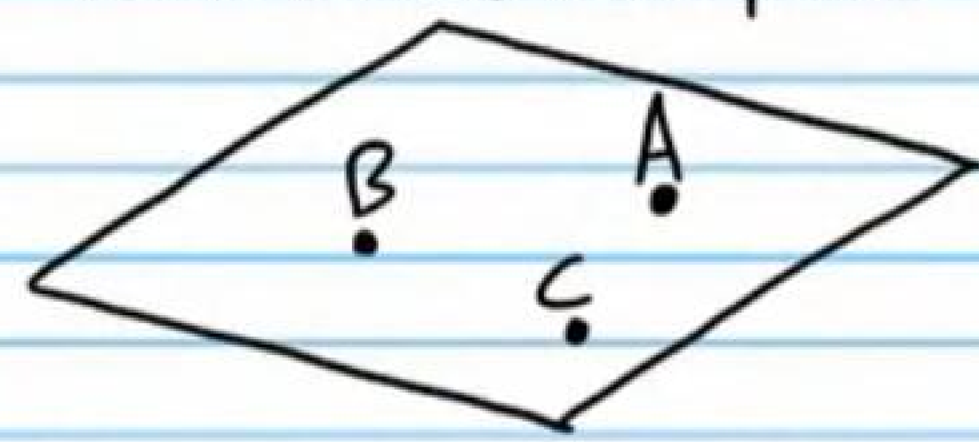
$\overleftrightarrow{AB}$ is the line of intersection.

Collinear Points

Points on the same line.

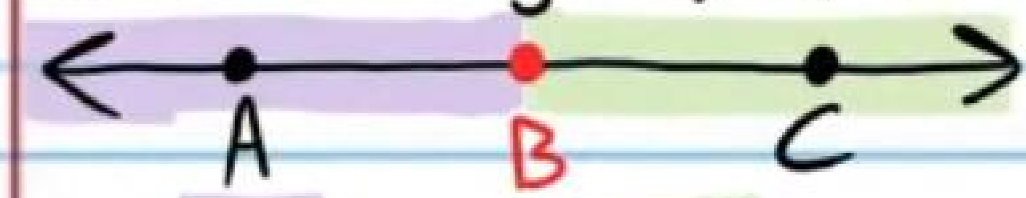
Coplanar Points

Points on the same plane



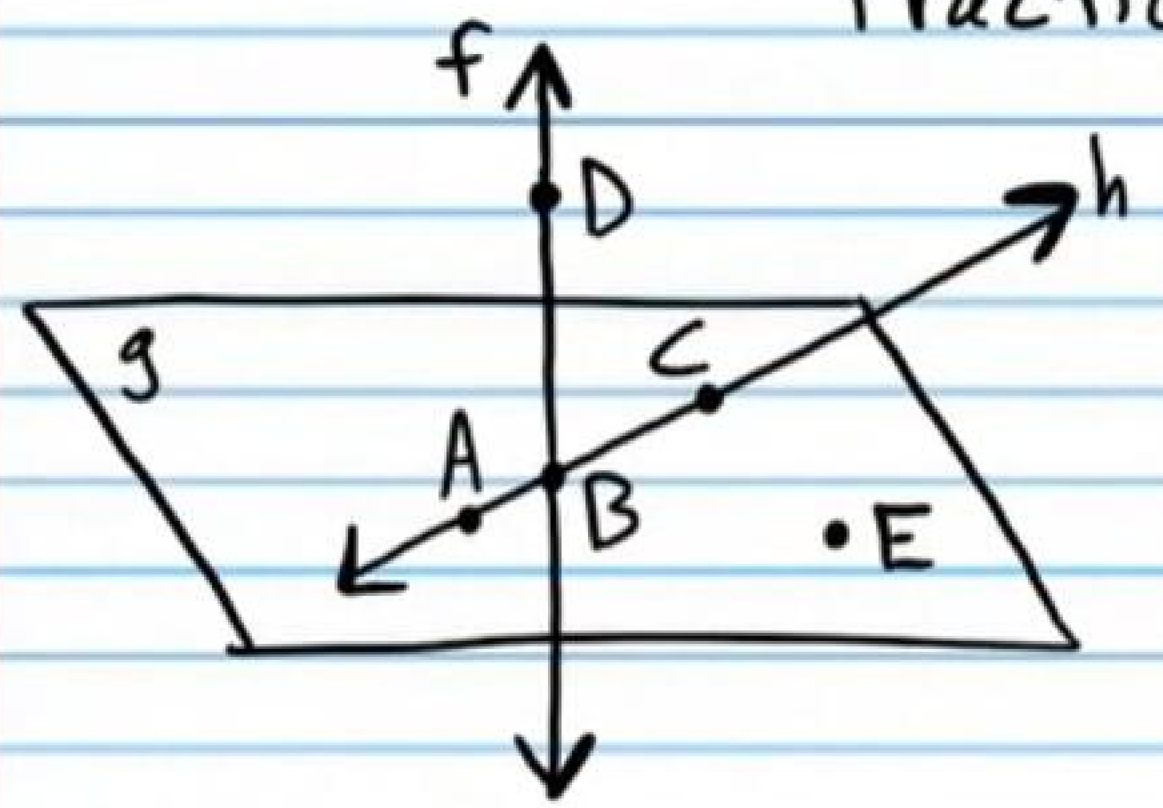
Opposite Rays

Rays with the same endpoint but extending in exactly opposite directions. Together, they create a line.



$\overrightarrow{BA}$ and $\overrightarrow{BC}$

Practice



Identify each of the following.

① $\overrightarrow{AB}$ $\boxed{\text{Ray}}$ ② E $\boxed{\text{Point}}$ ③ BCE $\boxed{\text{Plane}}$

④ $\overline{AC}$ $\boxed{\text{Segment}}$ ⑤ $\overleftrightarrow{BD}$ $\boxed{\text{Line}}$

⑥ Name three collinear points on the figure above. $\boxed{\text{A, B, and C}}$

⑦ Name four coplanar points. $\boxed{\text{A, B, C, and E}}$

⑧ What is the intersection of $\overleftrightarrow{DB}$ and $\overleftrightarrow{AC}$? $\boxed{\text{B}}$

⑨ Name a point that is not coplanar with ACE. $\boxed{\text{D}}$

⑩ Name a point that is not collinear with $\overleftrightarrow{AB}$. $\boxed{\text{D}}$

⑪ What is another name for $\overleftrightarrow{DB}$? $\boxed{\text{f}}$

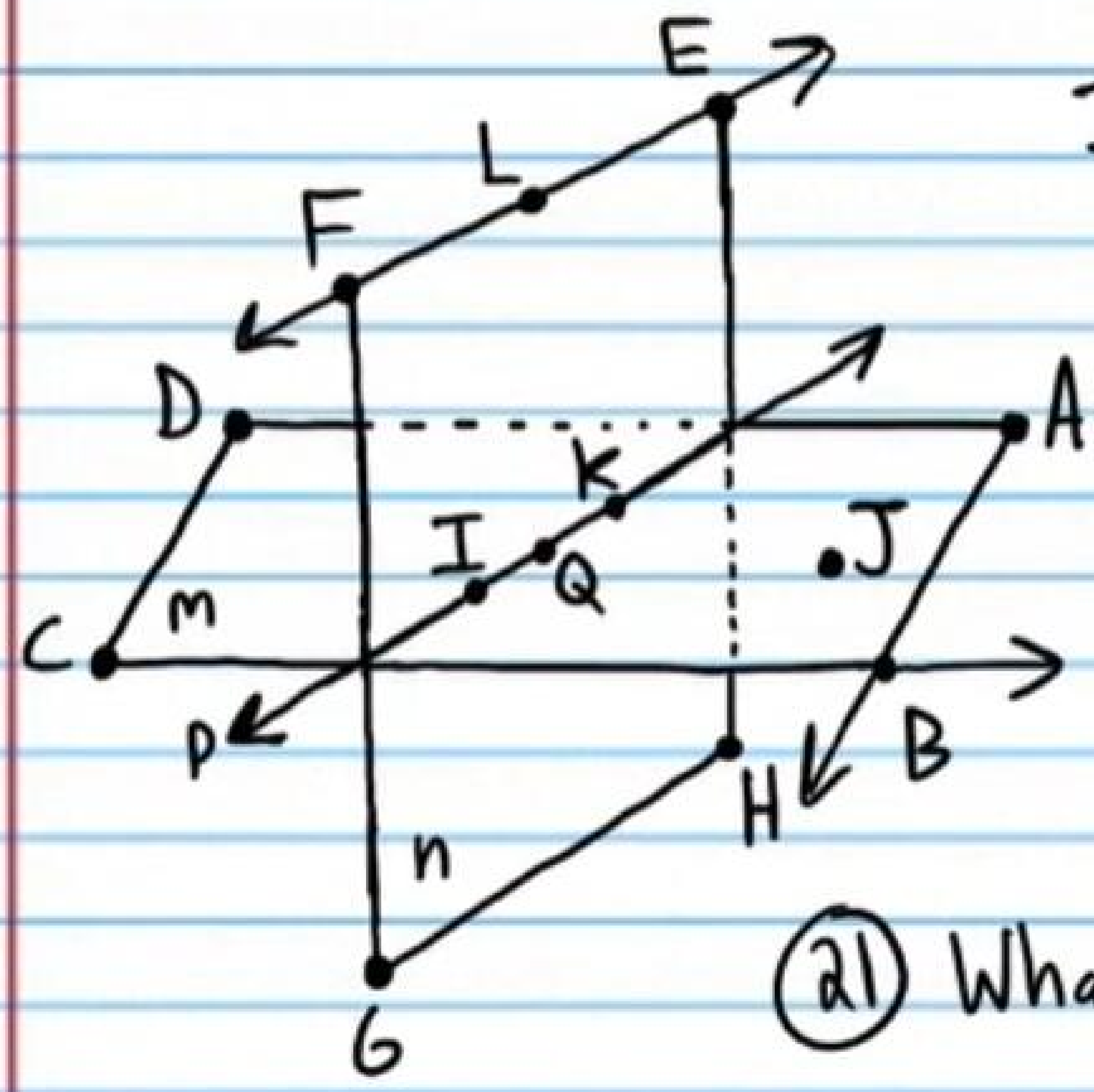
⑫ What is another name for CEB? $\boxed{\text{ACE}}$

⑬ Name two opposite rays.
$\boxed{\overrightarrow{BC} \text{ and } \overrightarrow{BA}}$

⑭ What is another name for $\overleftrightarrow{BC}$? $\boxed{\overleftrightarrow{AC}}$

⑮ Are A, E, and B collinear? $\boxed{\text{No}}$

More Practice



Identify each of the following.

⑯ $\overline{BC}$ $\boxed{\text{Segment}}$ ⑰ $\overleftrightarrow{IK}$ $\boxed{\text{Line}}$ ⑱ FGH $\boxed{\text{Plane}}$

⑲ C $\boxed{\text{Point}}$ ⑳ $\overrightarrow{KJ}$ $\boxed{\text{Ray}}$

㉑ What is another name for $\overleftrightarrow{IK}$? $\boxed{\overleftrightarrow{IQ}}$

㉒ What is another name for ABC? $\boxed{\text{IQJ}}$

㉓ What is the intersection of $\overrightarrow{CB}$ and $\overrightarrow{AB}$? $\boxed{\text{B}}$

㉔ Name a point that is not coplanar with ABD. $\boxed{\text{E}}$

㉕ Name a point that is not collinear with $\overleftrightarrow{FE}$. $\boxed{\text{B}}$

㉖ Name three collinear points. $\boxed{\text{F, L, and E}}$

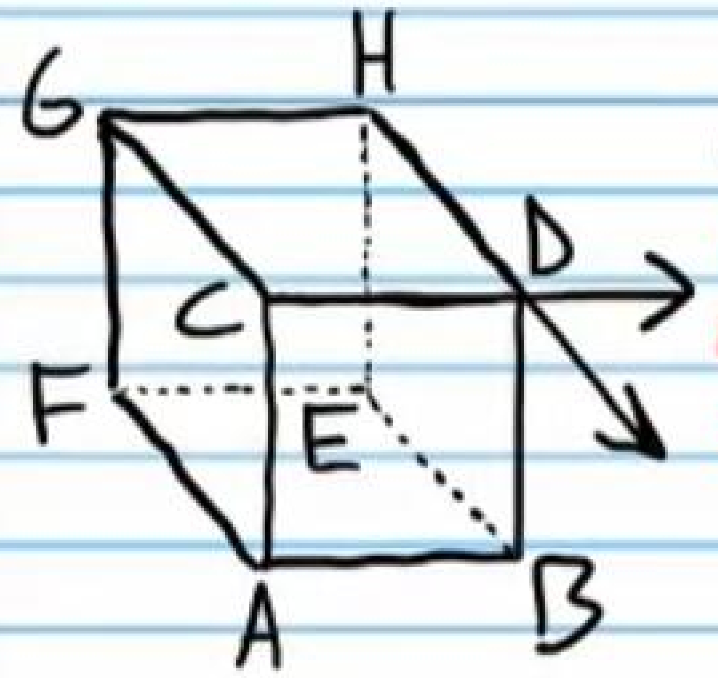
㉗ Name four coplanar points. $\boxed{\text{F, L, E, and Q}}$

㉘ Name two opposite rays. $\boxed{\overrightarrow{LF} \text{ and } \overrightarrow{LE}}$

㉙ What is another name for FEH? $\boxed{\text{FGH}}$

㉚ What is the intersection of n and m? $\boxed{\text{P}}$

More Practice



(31) Name the intersection of GFA and DBC.

Note: These are planes with finite length, which means that their intersection will be segments, not lines.

$\boxed{\overline{CA}}$

(32) Name three planes that intersect at B.

$\boxed{\text{CAB, FEB, HDB}}$

(33) Name the intersection of $\overrightarrow{CD}$ and $\overrightarrow{HD}$.

$\boxed{D}$

(34) Name the intersection of GHE and GCD.

$\boxed{\overline{GH}}$

(35) Are G, H, and E collinear?

$\boxed{\text{No}}$

(36) Are F, E, and A coplanar?

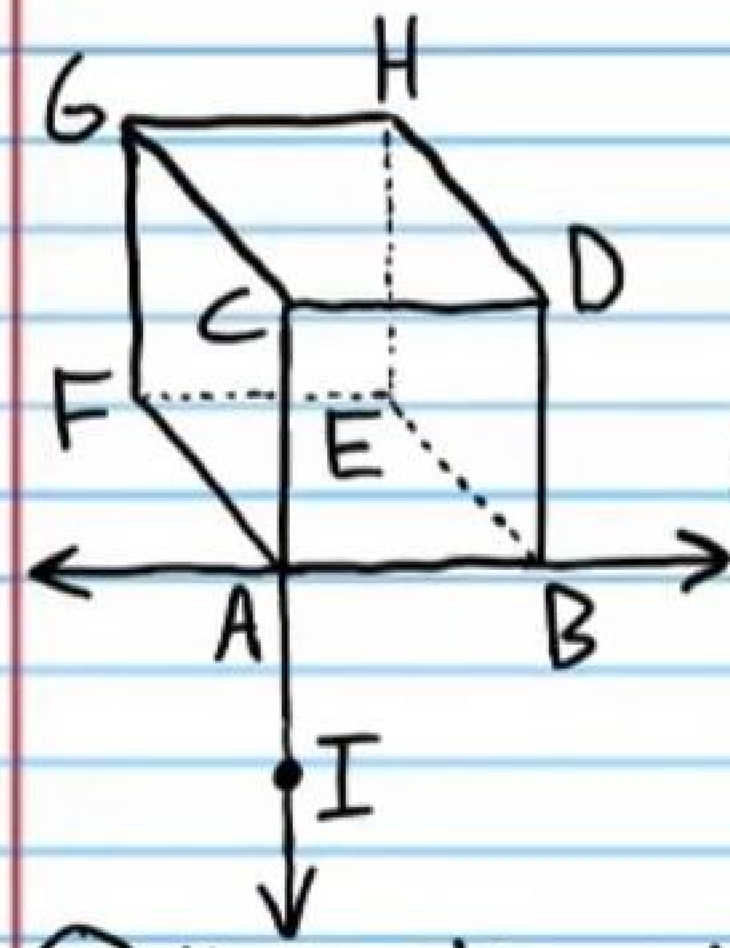
$\boxed{\text{Yes}}$

(37) Are A and G collinear? $\boxed{\text{Yes}}$

(38) Are A and H coplanar? $\boxed{\text{Yes}}$

(39) Are G, C, A, and B coplanar? $\boxed{\text{No}}$

More Practice



(40) Name three planes that intersect at G.

$\boxed{\text{GHD, FGC, GHE}}$

(41) Name the intersection of EHG and HDB.

Note: These are planes with finite length, which means that their intersection will be segments, not lines.

$\boxed{\overline{HE}}$

(42) Name the intersection of $\overleftrightarrow{AB}$ and $\overrightarrow{CA}$. $\boxed{A}$

(43) Name the intersection of FEB and ABD.

$\boxed{\overline{AB}}$

(44) Are G, C, and D collinear?

$\boxed{\text{No}}$

(45) Are C and D collinear?

$\boxed{\text{Yes}}$

(46) Are F, G, and A coplanar?

$\boxed{\text{Yes}}$

(47) Are F and A coplanar? $\boxed{\text{Yes}}$

(48) Are C, A, and I collinear? $\boxed{\text{Yes}}$

Postulate (also called an axiom): A statement that is accepted without an explanation or proof.

Theorem: A statement that requires an explanation or proof that uses definitions, postulates, other theorems, logic, etc.

Segments

If $\overline{AB}$ is a segment, then AB is defined as its length.

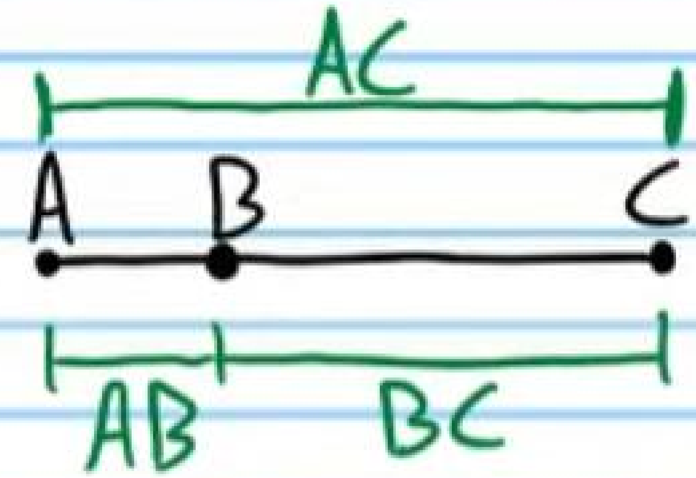
Congruence: Having the same size and shape (notated by $\cong$).

If two segments are congruent, then they have the equal lengths.

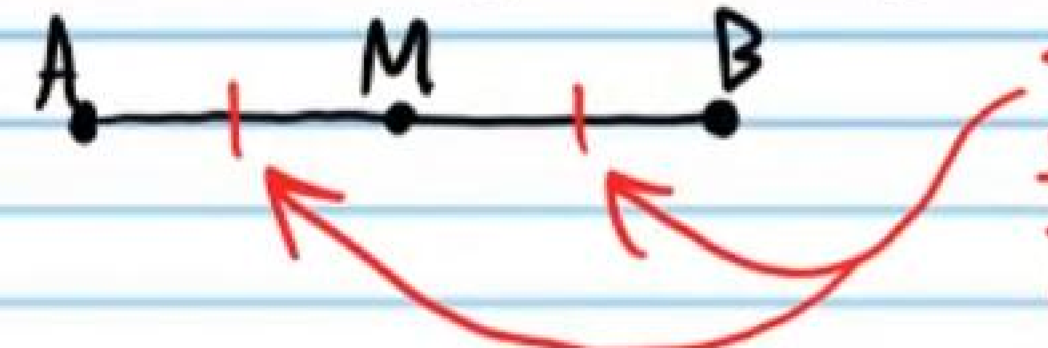
Example: If $\overline{AB} \cong \overline{CD}$, then $AB = CD$.

The Segment Addition Postulate

If A, B, and C are collinear, and B is between A and C, then $AB + BC = AC$.



Midpoint of a Segment: A point (e.g. M below) that is in the middle or center of a segment, dividing it into two congruent segments.

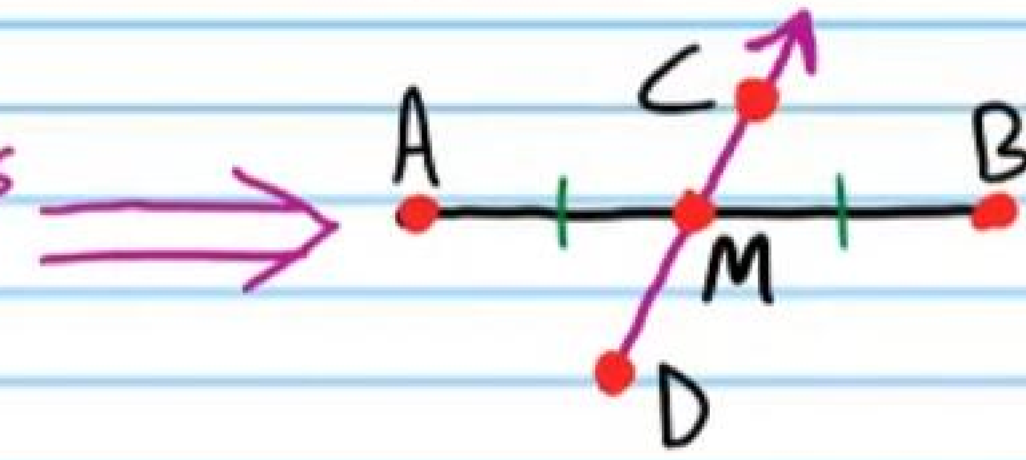


These red marks indicate that segments $\overline{AM}$ and $\overline{MB}$ are congruent.

Bisect: Divide into two equal parts.

Segment Bisector: A point, ray, line, segment, or plane that divides a segment into two congruent segments by intersecting at its midpoint.

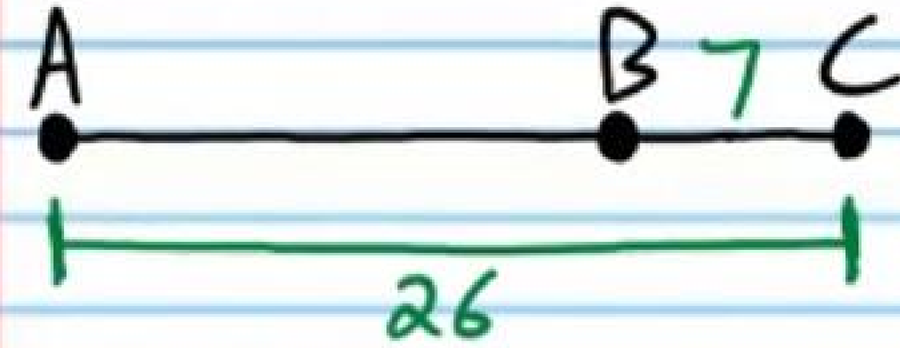
Using a ray as an example.



$\overrightarrow{DC}$ is a segment bisector.

Practice

(49) Find AB.

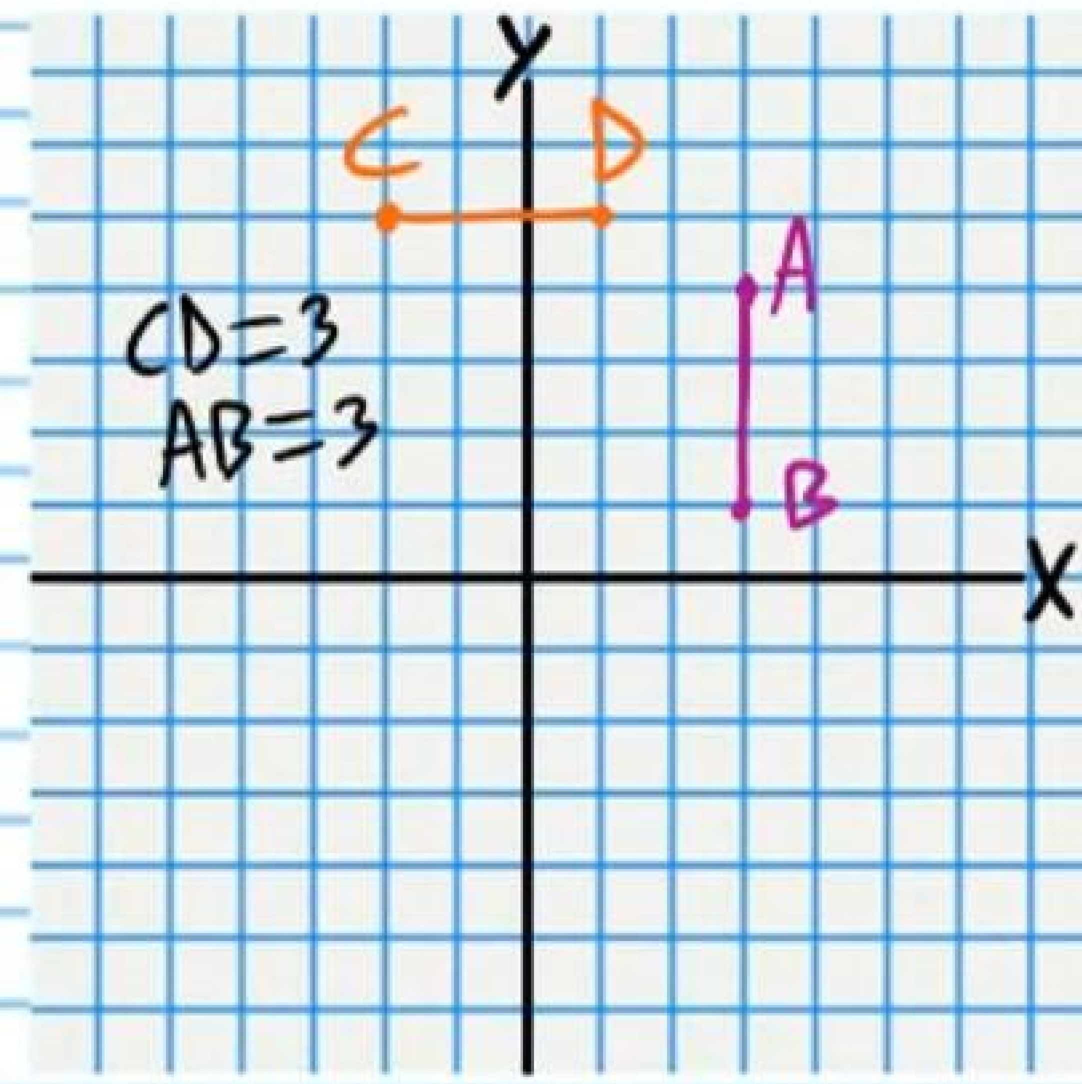


$AC = AB + BC$

$26 = AB + 7$

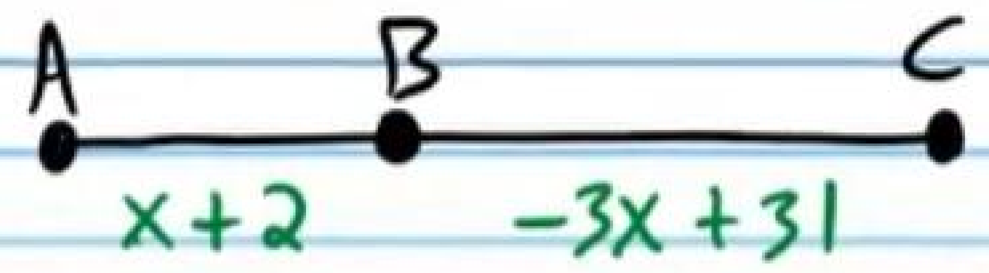
$\boxed{19 = AB}$

(50) If $A(3,4)$ and $B(3,1)$ are the endpoints of a segment, and $C(-2,5)$ and $D(1,5)$ are the endpoints of a different segment, are $\overline{AB}$ and $\overline{CD}$ congruent?



$\boxed{\text{Yes}}$

51. If $AC = 23$, find AB.



I) $AC = AB + BC$

$23 = x + 2 + (-3x + 31)$

$23 = -2x + 33$

$-10 = -2x$

$x = 5$

II) $AB = x + 2$

$AB = 5 + 2$

$\boxed{AB = 7}$

52. If M is the midpoint of $\overline{AB}$, find AB.

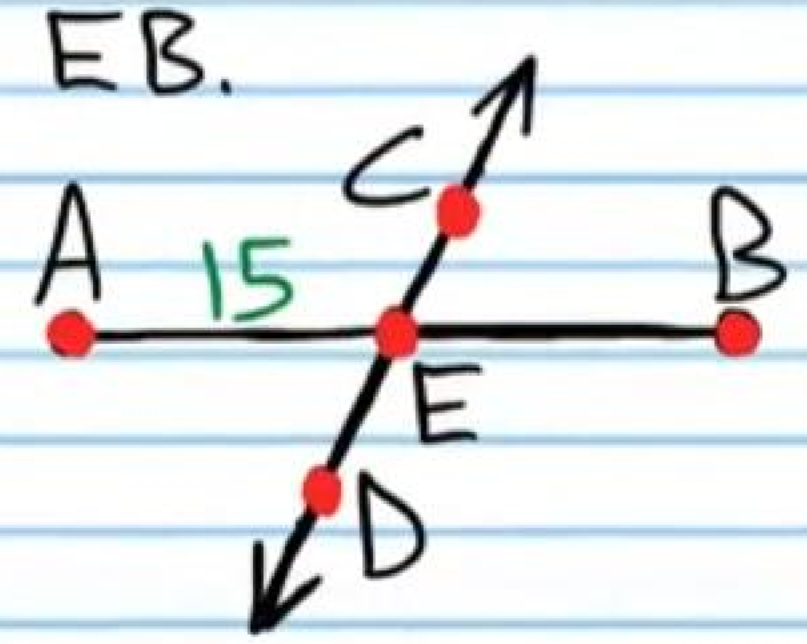
I) $\overline{AM} \cong \overline{MB}$

$AM = MB$

$9 = MB$

II) $AB = AM + MB = 9 + 9 = \boxed{18}$

53. If $\overleftrightarrow{CD}$ bisects $\overline{AB}$, find EB.

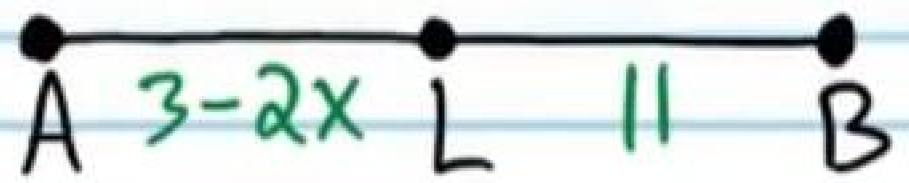


$\overline{AE} \cong \overline{EB}$

$AE = EB$

$\boxed{15 = EB}$

54. If L bisects $\overline{AB}$, find x.



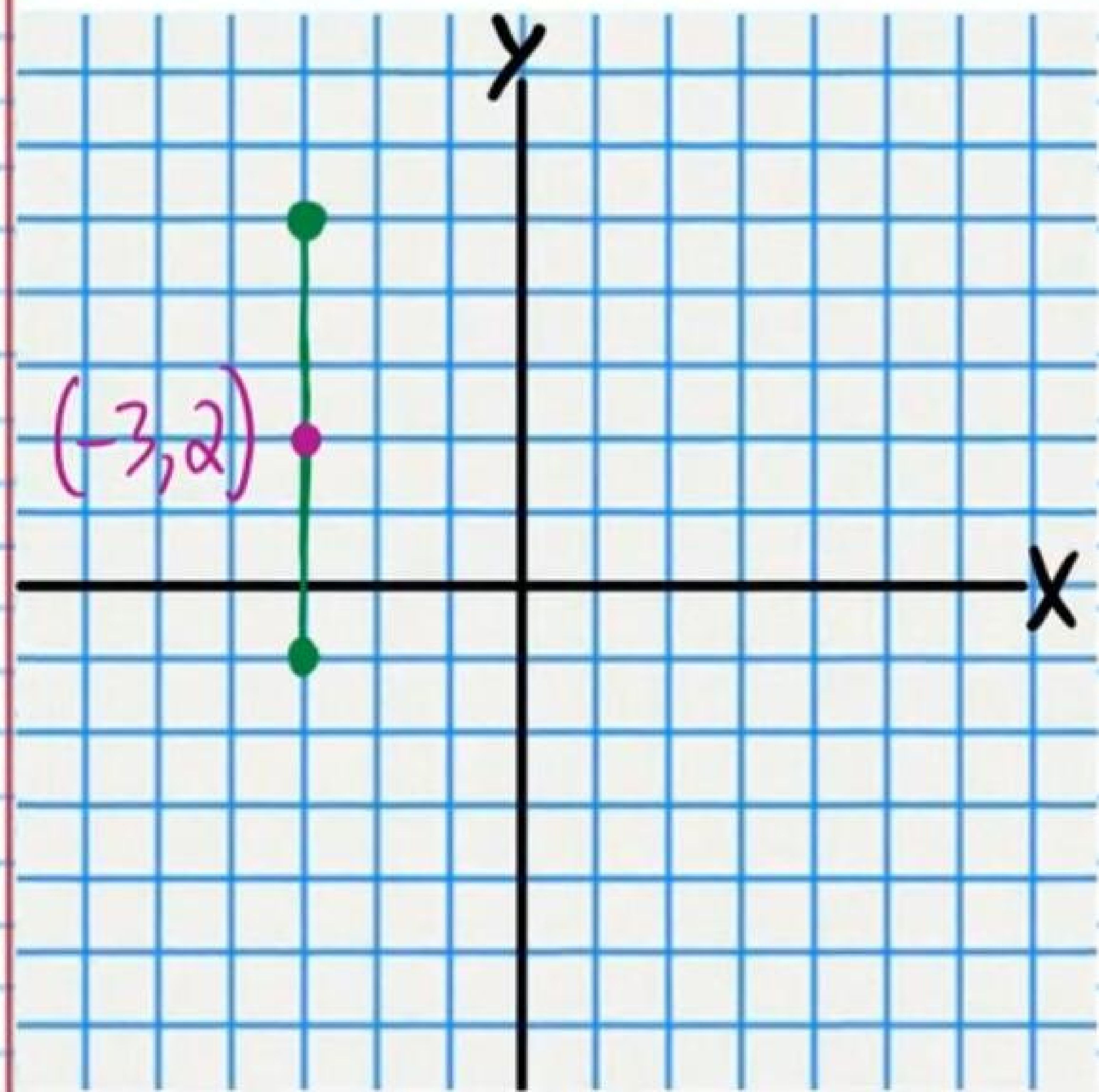
$\overline{AL} \cong \overline{LB}$

$AL = LB$

$3 - 2x = 11$

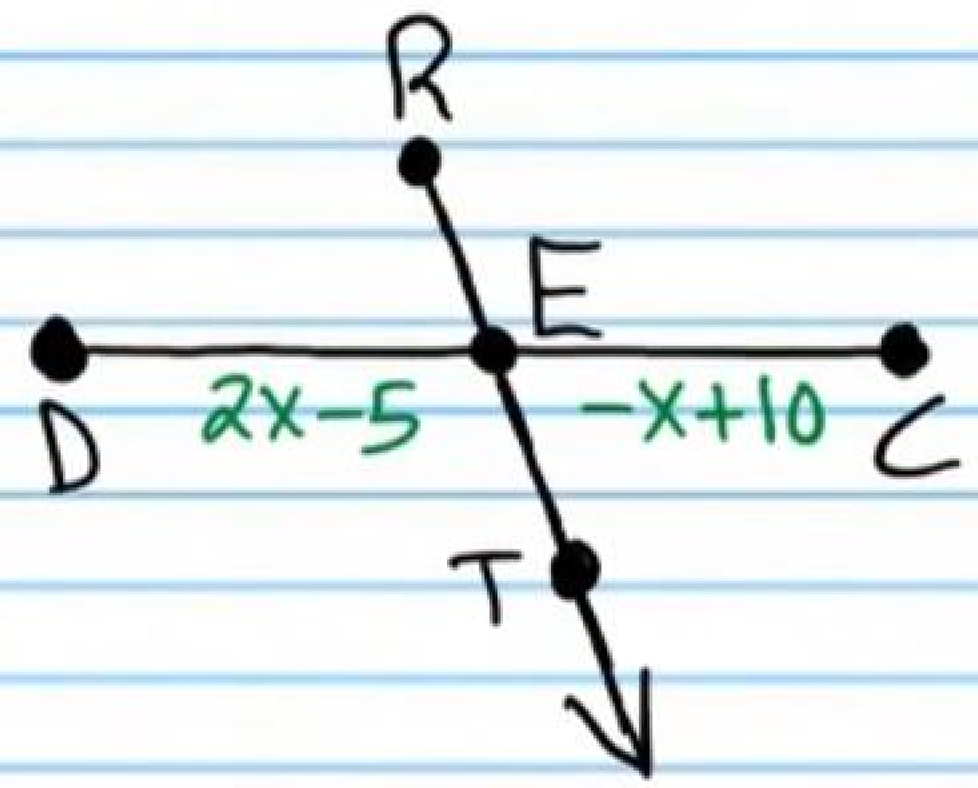
$\boxed{x = -4}$

55. Find the midpoint of the segment with endpoints $A(-3,-1)$ and $B(-3,5)$.



$\boxed{(-3,2)}$

(56) If $\overrightarrow{RT}$ bisects $\overline{DC}$, find DC.



I) $\overline{DE} \cong \overline{EC}$

$$DE = EC$$

$$2x-5 = -x+10$$

$$3x = 15$$

$$x = 5$$

II) $DC = DE + EC$

$$= 2x-5+(-x+10)$$

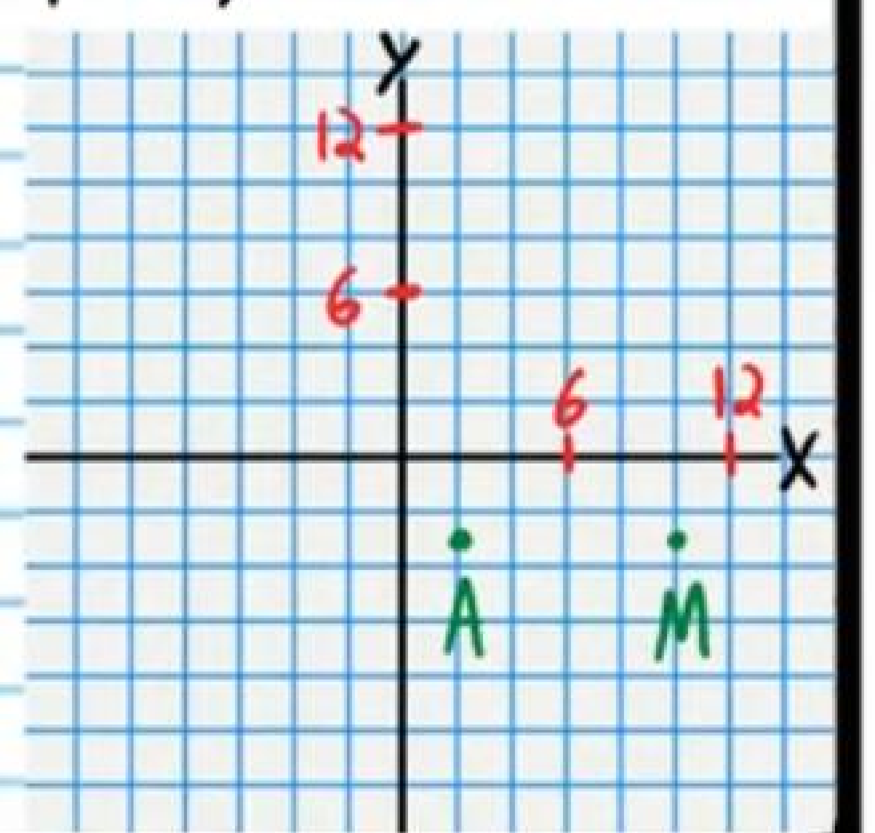
$$= x+5$$

$$= 5+5$$

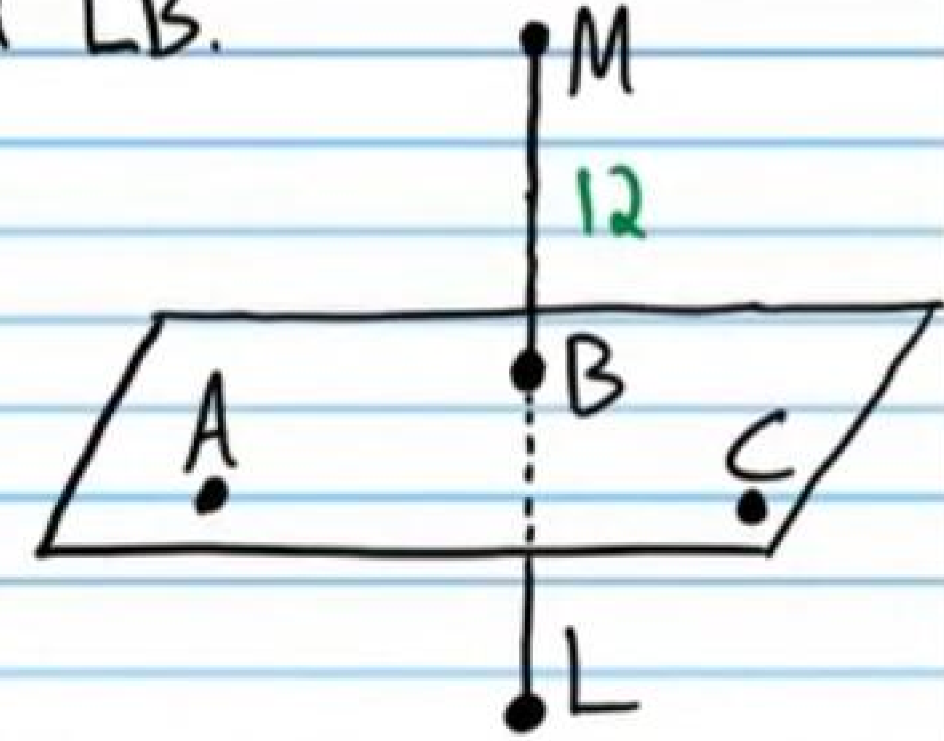
$$= \boxed{10}$$

(57) If A(2,−3) is an endpoint of the segment $\overline{AB}$, and M(10,−3) is the segment's midpoint, find the coordinates of B.

$\boxed{B(18,-3)}$



(58) If ABC bisects $\overline{LM}$, find LB.

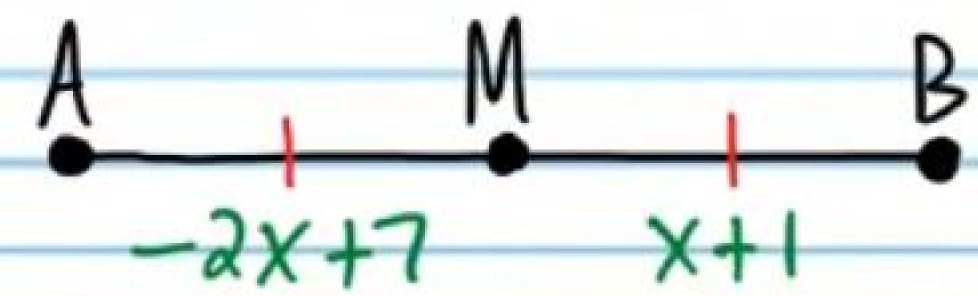


$\overline{MB} \cong \overline{BL}$

$$MB = BL$$

$$\boxed{12 = BL}$$

(59) If M is the midpoint of $\overline{AB}$, find the length of AM if $AM = -2x+7$ and $MB = x+1$.



I) $\overline{AM} \cong \overline{MB}$

$$AM = MB$$

$$-2x+7 = x+1$$

$$6 = 3x$$

$$2 = x$$

II) $AM = -2x+7$

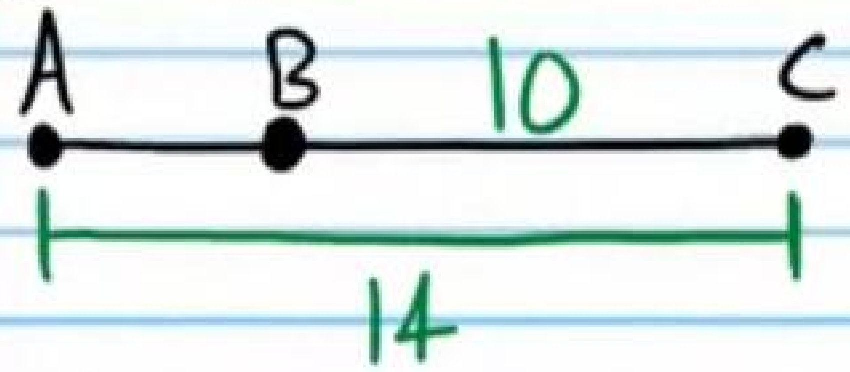
$$= -2(2)+7$$

$$= -4+7$$

$$= \boxed{3}$$

More Practice

(60) Find AB.

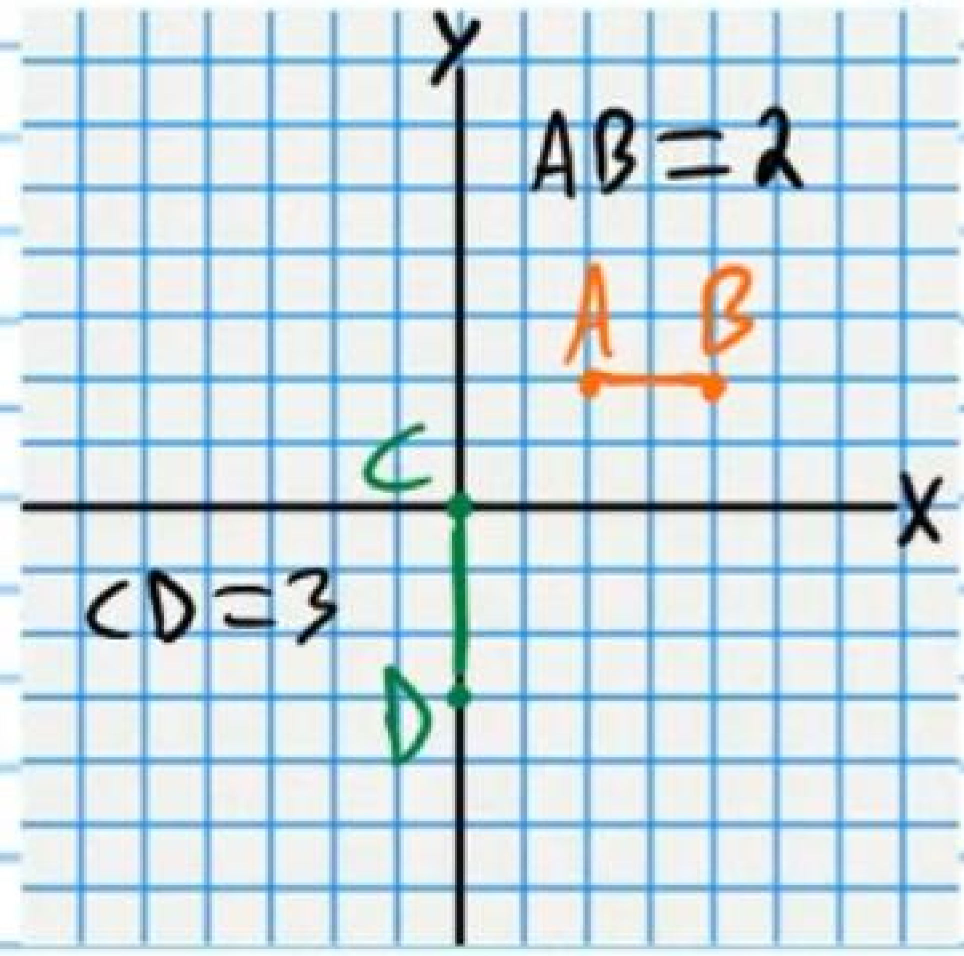


$AC = AB + BC$

$14 = AB + 10$

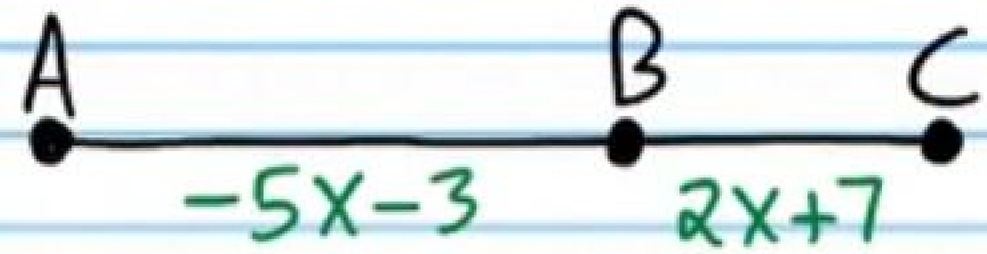
$\boxed{4 = AB}$

(61) If $A(2,2)$ and $B(4,2)$ are the endpoints of a segment, and $C(0,0)$ and $D(0,-3)$ are the endpoints of a different segment, are $\overline{AB}$ and $\overline{CD}$ congruent?



$\boxed{\text{No}}$

(62) If $AC = 10$, find BC and AB.



I) $AC = AB + BC$

$10 = -5x - 3 + 2x + 7$

$10 = -3x + 4$

$6 = -3x$

$-2 = x$

II) $BC = 2x + 7 = 2(-2) + 7 = \boxed{3}$

III) $AB = -5x - 3$

$= -5(-2) - 3$

$= 10 - 3$

$= \boxed{7}$

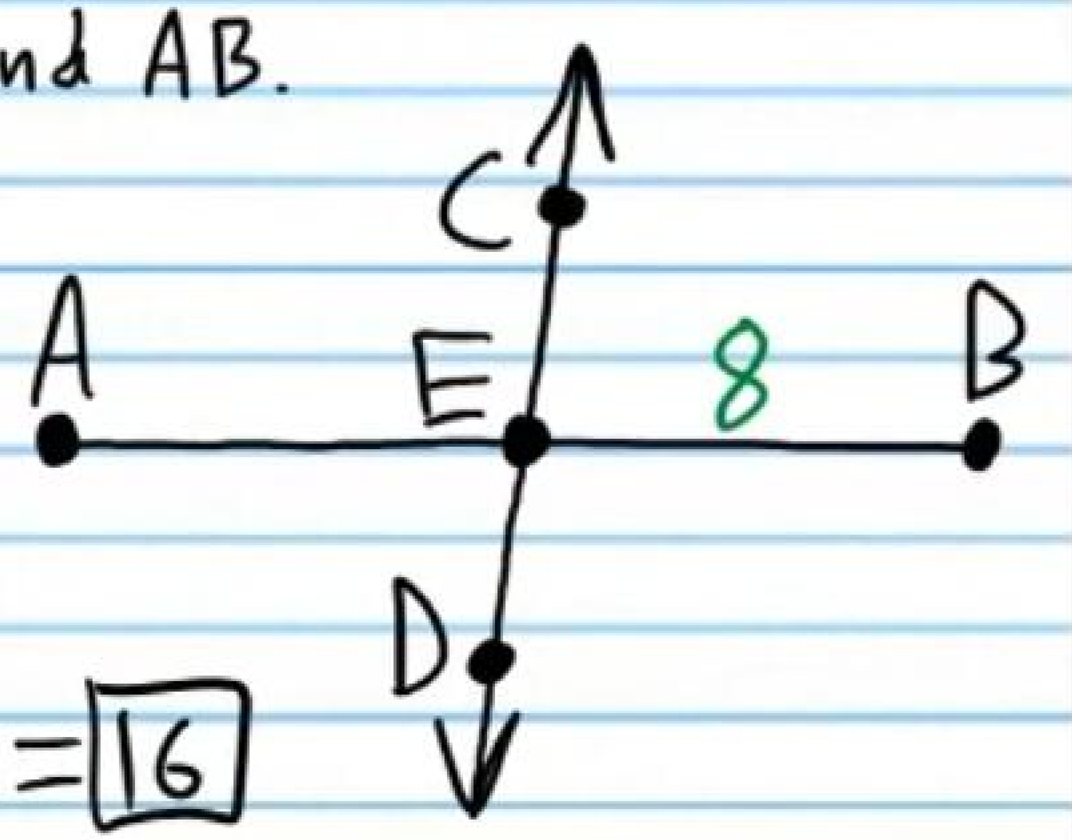
(63) If M is the midpoint of $\overline{AB}$, find AM.

$\overline{AM} \cong \overline{MB}$

$AM = MB$

$\boxed{AM = 12}$

(64) If $\overleftrightarrow{CD}$ bisects $\overline{AB}$, find AB.



I) $\overline{AE} \cong \overline{EB}$

$AE = EB$

$AE = 8$

II) $AB = AE + EB = 8 + 8 = \boxed{16}$

(65) If L bisects $\overline{AB}$, find x.

$\overline{AL} \cong \overline{LB}$

$AL = LB$

$10x + 1 = \frac{3x}{2} + 2$

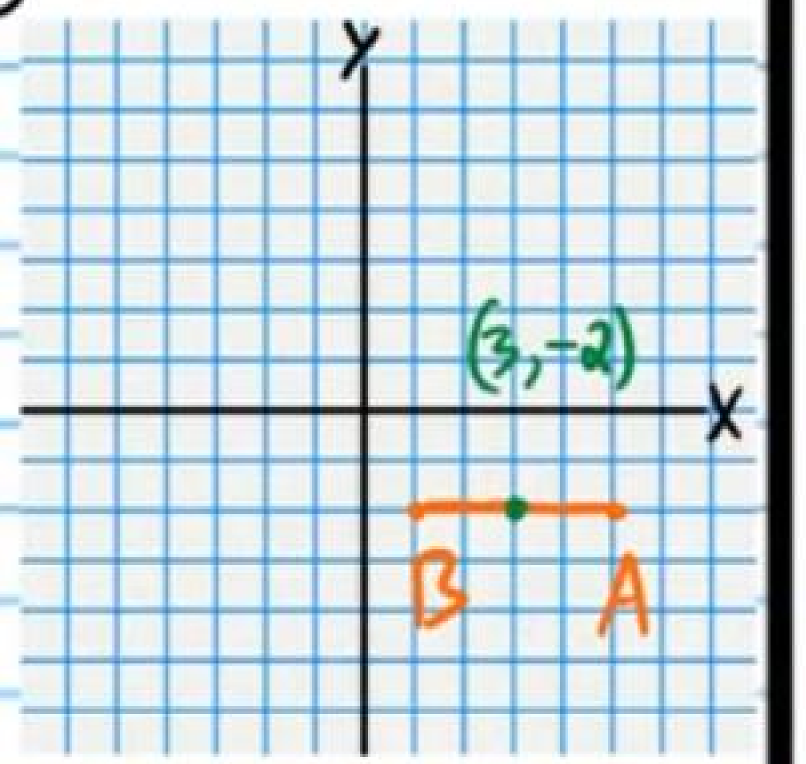
$2(10x + 1) = 2\left(\frac{3x}{2} + 2\right)$

$20x + 2 = 3x + 4$

$17x = 2$

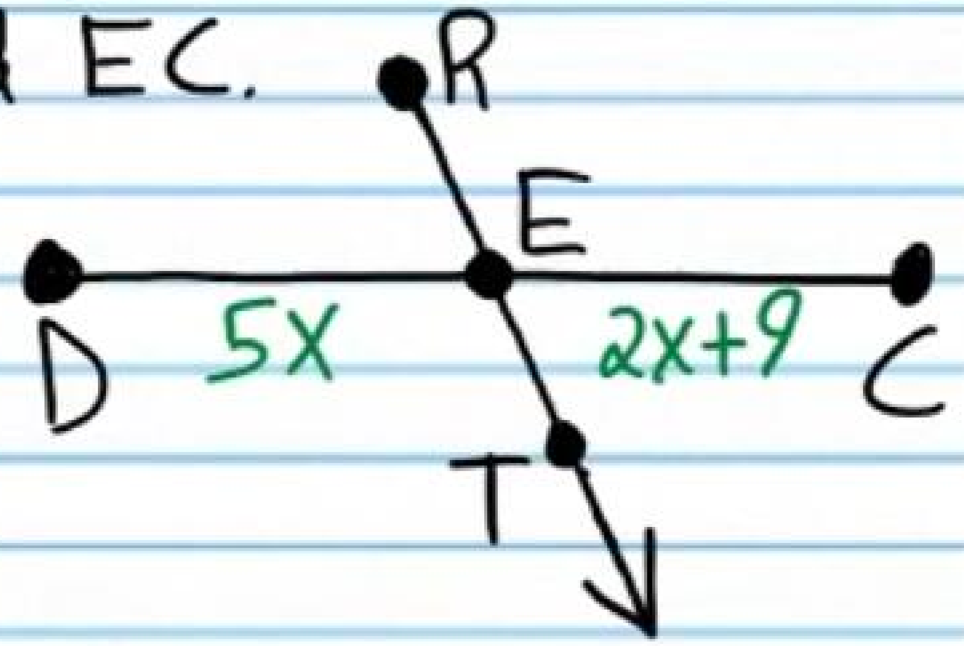
$\boxed{x = \frac{2}{17}}$

(66) Find the midpoint of the segment with endpoints A(5,−2) and B(1,−2).



$\boxed{(3,-2)}$

(67) If $\overrightarrow{RT}$ bisects $\overline{DC}$, Find EC.



I) $\overline{DE} \cong \overline{EC}$

$DE = EC$

$5x = 2x+9$

$3x = 9$

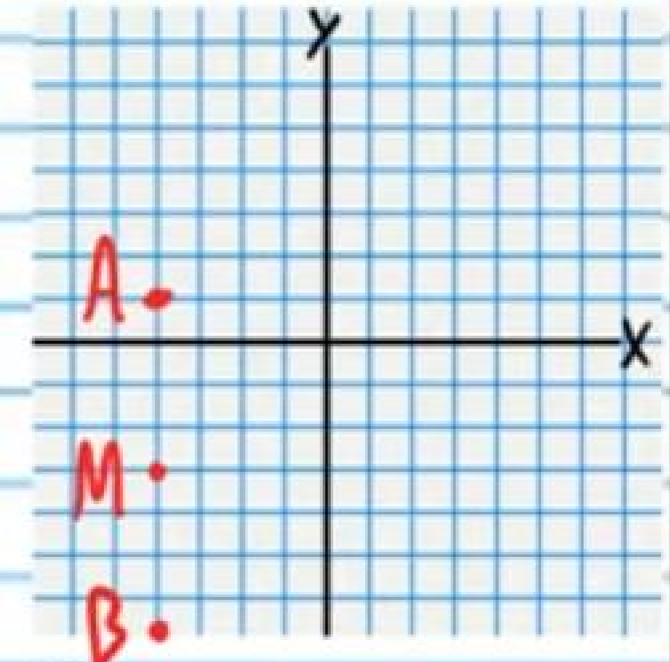
$x = 3$

II) $EC = 2x+9$

$EC = 2(3)+9$

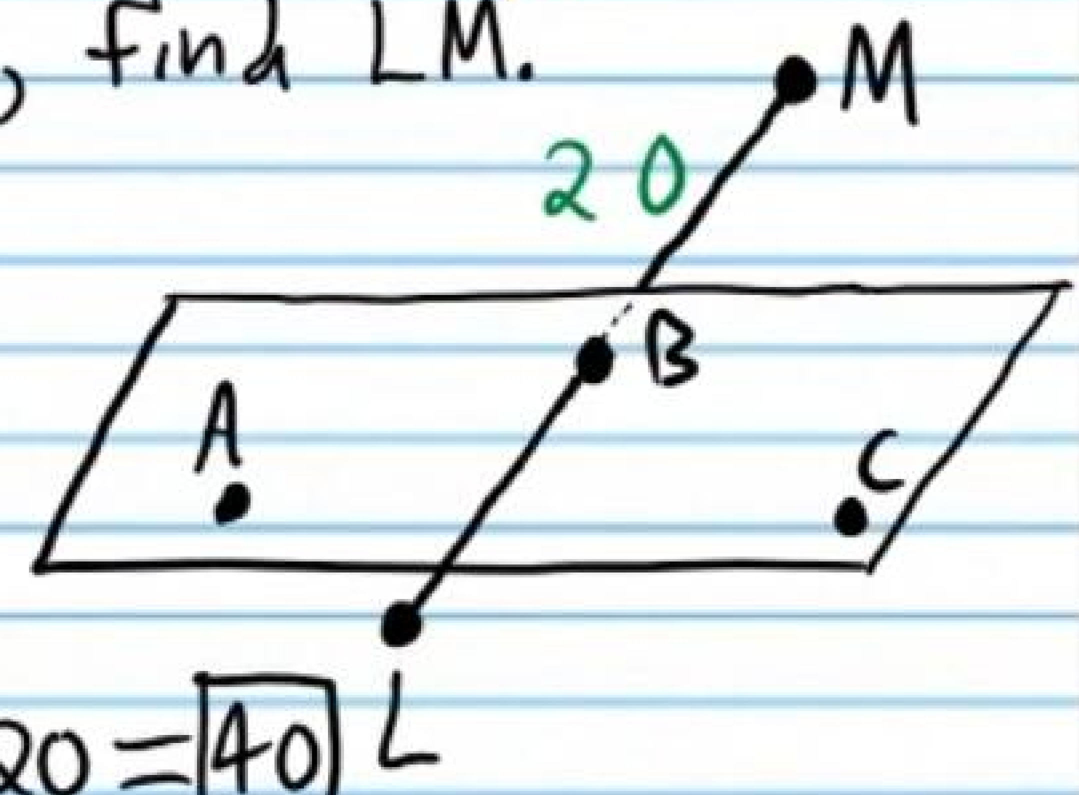
$EC = \boxed{15}$

(68) If A(−4,1) is an endpoint of the segment $\overline{AB}$, and M(−4,−3) is the segment's midpoint, find the coordinates of B.



$\boxed{B(-4,-7)}$

(69) If ABC bisects $\overline{LM}$, find LM.



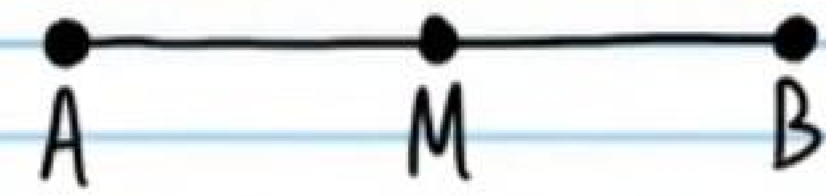
I) $\overline{LB} \cong \overline{BM}$

$LB = BM$

$LB = 20$

II) $LM = LB + BM = 20 + 20 = \boxed{40}$

(70) If M is the midpoint of $\overline{AB}$, find the length of AB if $AM = 6x-11$ and $MB = 10x-51$.



I) $\overline{AM} \cong \overline{MB}$

$AM = MB$

$6x-11 = 10x-51$

$40 = 4x$

$10 = x$

II) $AB = AM + MB$

$AB = 6x-11 + 10x-51$

$AB = 16x-62$

$AB = 16(10)-62$

$AB = \boxed{98}$